1. Pseudocode of Intelligent Hospital Emergency Management Using Data Structure

Hospital Management System Pseudocode

```
DECLARE Critical Patient Priority Queue
DECLARE Normal Patient Queue
DECLARE Doctor List/Collection
DECLARE Historical Records List
BEGIN
CALL Open Login Page
READ Password
 IF  Password is Correct   THEN
    REPEAT
       DISPLAY  Main Menu (1- Patient Management, 2- Doctor Management, 0- Logout)
         READ  User Choice

        WHEN  Patient Management (Choice 1)
           DISPLAY   Patient Menu (1- Add, 2- Display, 3- Remove, 4- Replace/Change, 0- Back)
           READ   Patient Choice

         IF  Add Patient  THEN
             READ   Patient Data
            IF  Status is Critical  THEN
                ADD  Patient to Critical Priority Queue
             ELSE
                ADD  Patient to Normal Queue
             END IF

          ELSE IF  Display Patients  THEN
            DISPLAY  All Patients by Priority

        ELSE IF  Remove/Register  THEN
           READ  Patient ID
           IF  ID Found  THEN
                REMOVE  Patient from Queue
           ELSE
                PRINT  "Patient Not Found"
            END IF
         END IF
 WHEN  Doctor Management (Choice 2)
         DISPLAY  Doctor Menu (1- Add, 2- Display, 3- Remove, 0- Back)
          READDoctor Choice

        IF  Add Doctor  THEN
           READ  Doctor Data and  ADD  to Doctor List
          ELSE IF  Display Doctors  THEN
             DISPLAY  All Registered Doctors
          ELSE IF  Remove Doctor  THEN
            READ  Doctor ID and   REMOVE  if found
          END IF
```

```
 WHEN  System Auto-Execution (Every Clock Tick)
        FOR EACH Doctor
           IF Doctor is Free THEN
              IF  Critical Patient Available  AND  Specialty Matches  THEN
                 ASSIGN  Critical Patient to Doctor
                 SET  Doctor Status to Busy
             ELSE IF  Normal Patient Available  THEN
                ASSIGN  Normal Patient to Doctor
                 SET Doctor Status to Busy
             END IF
            ELSE IF  Treatment is Done  THEN
             SET Doctor Status to Free
              SAVE  Action to History
            END IF
          END FOR
UNTIL User Choice is Logout (0)
 END IF
END
```